

Module 1

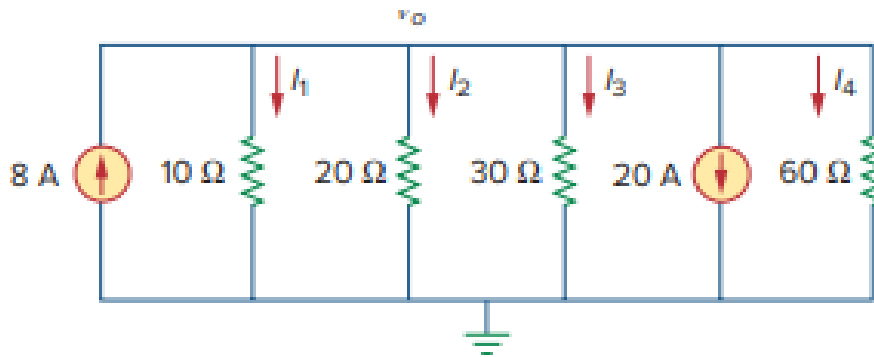
DC Circuit Analysis and Theorems

Dr. Elizabeth Rufus

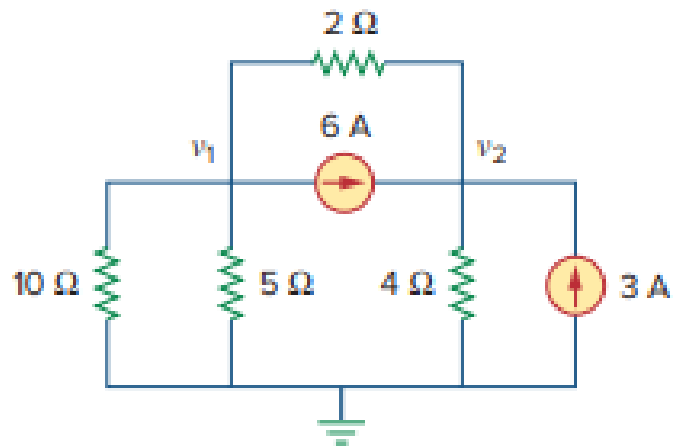
Professor, School of Electronics Engineering

Tutorial

1. Find the currents I_1 through I_4 and the voltage v_o in the circuit using Nodal analysis method.

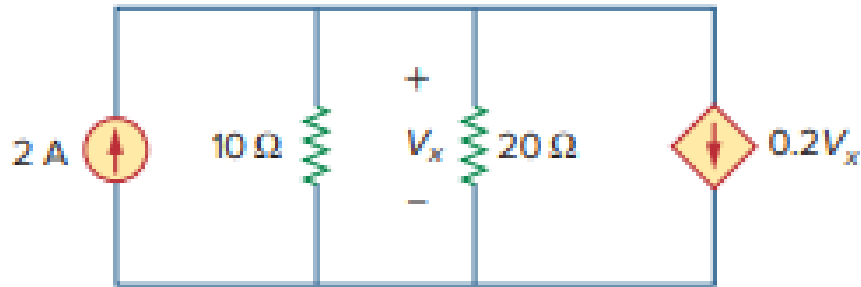


2. Find the voltages v_1 and v_2 using the nodal analysis method.

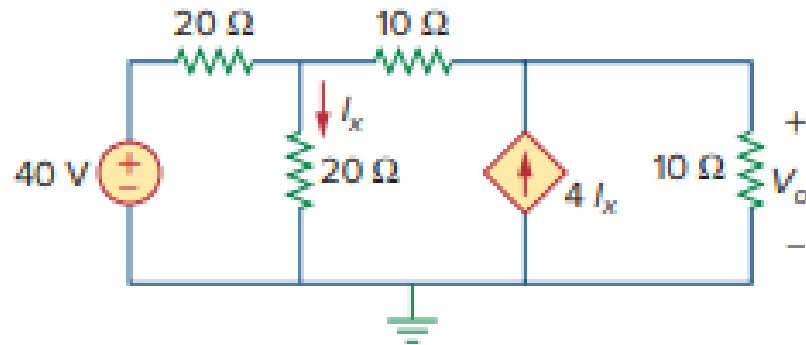


Tutorial

3. Apply nodal analysis to solve for V_x in the circuit.

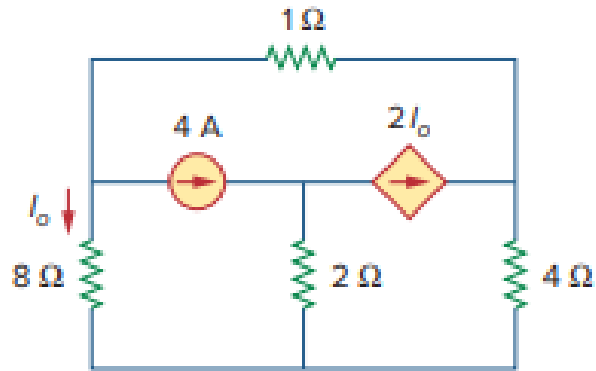


4. Using nodal analysis, determine V_o in the circuit.

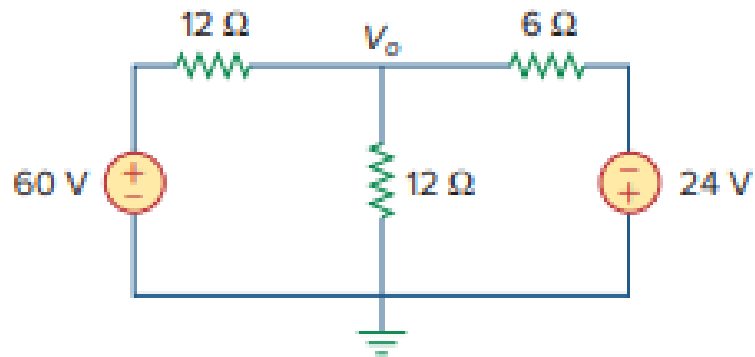


Tutorial

5. Find I_o in the circuit using Nodal analysis method.

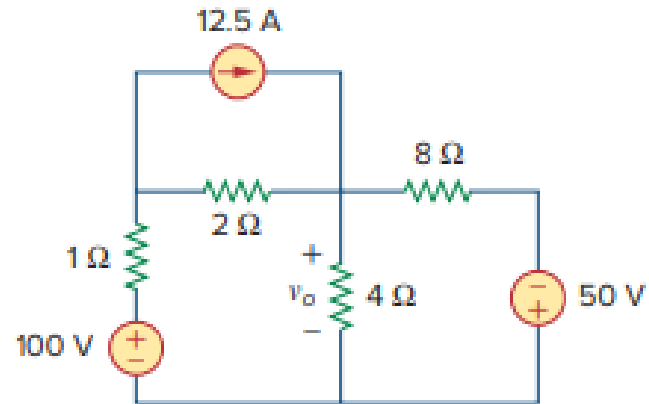


6. Find V_o and the power dissipated in all the resistors in the circuit. Use Nodal analysis method.



Tutorial

7. Using nodal analysis, find v_o in the circuit.



8. Determine voltages v_1 through v_3 in the circuit using nodal analysis.

